

WORKED EXAMPLE

Two inputs, one prediction

Each example has two measured features. It still has one target.

Our study data are $(x_1, x_2, y) = (1, 0, 2), (0, 1, 1), (1, 1, 3)$.

$$h_{\theta}(x) = \theta^T x = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

Lecture 1, page 43. The dot product multiplies matching components and adds them. T means transpose.

$$x_0 = 1, \quad \theta_0 = b, \quad \theta_1 = w_1, \quad \theta_2 = w_2$$

$$h(x) = b + w_1 x_1 + w_2 x_2$$

$x_0 = 1$ is bookkeeping for the bias, not a third measured feature. Parameter order throughout: (b, w_1, w_2) .

WORKED EXAMPLE

Start with three zero parameters

Feature subscripts choose columns; example superscripts choose rows.

$$(b, w_1, w_2) = (0, 0, 0), \quad \alpha = 0.3, \quad m = 3$$

$$h_1 = 0 + 0(1) + 0(0) = 0, \quad r_1 = 0 - 2 = -2$$

$$h_2 = 0 + 0(0) + 0(1) = 0, \quad r_2 = 0 - 1 = -1$$

$$h_3 = 0 + 0(1) + 0(1) = 0, \quad r_3 = 0 - 3 = -3$$

$$J = \frac{(-2)^2 + (-1)^2 + (-3)^2}{2(3)} = \frac{14}{6} \approx 2.333333$$

Residual r is prediction minus target. J is half-MSE, matching Lecture 1.

WORKED EXAMPLE

One gradient for every parameter

Differentiate the same loss with respect to each parameter separately.

$$\frac{\partial r_i}{\partial b} = 1, \quad \frac{\partial r_i}{\partial w_1} = x_1^{(i)}, \quad \frac{\partial r_i}{\partial w_2} = x_2^{(i)}$$

$$J = \frac{1}{2m} \sum_i r_i^2 \implies \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_i r_i x_j^{(i)}$$

The square derivative contributes $2r$; 2 cancels $1/2$. Multiply by the relevant input through the chain rule.

$$g_b = \frac{\sum_i r_i}{m}, \quad g_1 = \frac{\sum_i r_i x_1^{(i)}}{m}, \quad g_2 = \frac{\sum_i r_i x_2^{(i)}}{m}$$

Lecture 1, page 44: bias is the same rule with $x_0 = 1$. Average each column once.

WORKED EXAMPLE

First step: calculate all contributions

Each row contributes (residual, residual times x1, residual times x2).

$$i = 1: \quad (-2, (-2)(1), (-2)(0)) = (-2, -2, 0)$$

$$i = 2: \quad (-1, (-1)(0), (-1)(1)) = (-1, 0, -1)$$

$$i = 3: \quad (-3, (-3)(1), (-3)(1)) = (-3, -3, -3)$$

$$g_b = \frac{-2 - 1 - 3}{3} = -2$$

$$g_1 = \frac{-2 + 0 - 3}{3} = -\frac{5}{3}, \quad g_2 = \frac{0 - 1 - 3}{3} = -\frac{4}{3}$$

A zero input makes that example contribute zero to that weight gradient, even if its prediction is wrong.

WORKED EXAMPLE

First simultaneous update

Use the same old parameter vector for all three right-hand sides.

$$b_{\text{new}} = 0 - 0.3(-2) = 0.6$$

$$w_{1,\text{new}} = 0 - 0.3(-5/3) = 0.5$$

$$w_{2,\text{new}} = 0 - 0.3(-4/3) = 0.4$$

$$(b, w_1, w_2) = (0.6, 0.5, 0.4)$$

Store the three new values, then replace the old ones together. One pass over all three examples produced one full-batch update.

WORKED EXAMPLE

Recompute after the first update

The second iteration must start with fresh predictions and residuals.

$$h_1 = 0.6 + 0.5(1) + 0.4(0) = 1.1, \quad r_1 = 1.1 - 2 = -0.9$$

$$h_2 = 0.6 + 0.5(0) + 0.4(1) = 1.0, \quad r_2 = 1 - 1 = 0$$

$$h_3 = 0.6 + 0.5(1) + 0.4(1) = 1.5, \quad r_3 = 1.5 - 3 = -1.5$$

$$J = \frac{(-0.9)^2 + 0^2 + (-1.5)^2}{6} = \frac{0.81 + 0 + 2.25}{6} = 0.51$$

The cost fell from 2.333333 to 0.51. Do not reuse the initial residuals.

WORKED EXAMPLE

Second step: contributions and means

Again, the order in every triple is (bias, weight 1, weight 2).

$$i = 1: \quad (-0.9, (-0.9)(1), (-0.9)(0)) = (-0.9, -0.9, 0)$$

$$i = 2: \quad (0, 0(0), 0(1)) = (0, 0, 0)$$

$$i = 3: \quad (-1.5, (-1.5)(1), (-1.5)(1)) = (-1.5, -1.5, -1.5)$$

$$g_b = \frac{-0.9 + 0 - 1.5}{3} = -0.8$$

$$g_1 = \frac{-0.9 + 0 - 1.5}{3} = -0.8, \quad g_2 = \frac{0 + 0 - 1.5}{3} = -0.5$$

The second example is exactly predicted, so its current contribution is zero.

WORKED EXAMPLE

Second update and new predictions

Each parameter has its own gradient, but the learning rate is shared.

$$b = 0.6 - 0.3(-0.8) = 0.84$$

$$w_1 = 0.5 - 0.3(-0.8) = 0.74, \quad w_2 = 0.4 - 0.3(-0.5) = 0.55$$

$$h_1 = 0.84 + 0.74(1) + 0.55(0) = 1.58, \quad r_1 = 1.58 - 2 = -0.42$$

$$h_2 = 0.84 + 0.74(0) + 0.55(1) = 1.39, \quad r_2 = 1.39 - 1 = 0.39$$

$$h_3 = 0.84 + 0.74(1) + 0.55(1) = 2.13, \quad r_3 = 2.13 - 3 = -0.87$$

WORKED EXAMPLE

Verify the second cost

The middle example gets worse, but the average fit still improves.

$$J = \frac{(-0.42)^2 + (0.39)^2 + (-0.87)^2}{6}$$
$$= \frac{0.1764 + 0.1521 + 0.7569}{6} = \frac{1.0854}{6} = 0.1809$$

$$2.333333 \rightarrow 0.51 \rightarrow 0.1809$$

These are costs at initialization, after update 1, and after update 2.

Shared parameters create tradeoffs across examples. A smaller mean cost does not guarantee improvement for each point.

WORKED EXAMPLE

Connect this to Homework 1

The homework uses the same rules with six rows and learning rate 0.01.

Homework 1 starts $w_1 = w_2 = b = 0$. Its first data row is $(x_1, x_2, y) = (1, 2, 11)$.

$$h = 0 + 0(1) + 0(2) = 0, \quad r = 0 - 11 = -11$$

$$(r, rX_1, rX_2) = (-11, (-11)(1), (-11)(2)) = (-11, -11, -22)$$

This is only the FIRST ROW contribution, not the full gradient. Repeat for all six rows, sum each column and divide by 6.

Then update all three parameters using $\alpha = 0.01$. Our three-row toy data and $\alpha = 0.3$ are study choices, not the homework dataset.

Common mistakes: confusing a feature index with an example index; forgetting the bias; mixing parameter order.

INDEPENDENT PRACTICE

Your turn: a negative feature value

Use one example so you can check every multiplication by hand.

$$(x_1, x_2, y) = (2, -1, 4), \quad (b, w_1, w_2) = (1, 2, 3)$$

$$m = 1, \quad \alpha = 0.1, \quad J = \frac{1}{2}(h - y)^2$$

Calculate the prediction, signed residual, and three gradients.

Update all parameters simultaneously. Recompute the prediction and half-MSE.

Pay special attention to the sign of the weight 2 gradient.

WORKED ANSWER

Practice answer: one full update

The negative input reverses the sign of its weight contribution.

$$h = 1 + 2(2) + 3(-1) = 2, \quad r = 2 - 4 = -2$$

$$(g_b, g_1, g_2) = (-2, (-2)(2), (-2)(-1)) = (-2, -4, 2)$$

$$b = 1 - 0.1(-2) = 1.2, \quad w_1 = 2 - 0.1(-4) = 2.4$$

$$w_2 = 3 - 0.1(2) = 2.8$$

$$h_{\text{new}} = 1.2 + 2.4(2) + 2.8(-1) = 1.2 + 4.8 - 2.8 = 3.2$$

$$J_{\text{new}} = \frac{1}{2}(3.2 - 4)^2 = \frac{1}{2}(0.64) = 0.32 < 2 = J_{\text{old}}$$